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Summary

Staff-level developer-platform and reliability engineer with 12+ years building internal platforms, cloud infrastructure, and automation that improve developer productivity at scale. I lead ambiguous, cross-functional initiatives from architecture and RFC through production adoption, with a focus on paved paths, self-service, Kubernetes, platform security, and AI-enabled engineering workflows. I build systems that reduce friction for developers while preserving reliability, observability, and safe operational boundaries.

Skills

Languages: Go, Python, Bash
Developer Platforms: Kubernetes, Kubernetes controllers, Backstage, Helm, ArgoCD, Flux, Linux
Cloud & Infrastructure: AWS, GCP, Terraform, Ansible, Packer, multi-account and multi-region deployments
AI & Automation: AI-assisted developer tooling, agent workflows, LLM gateway architecture, internal AI plugins
Reliability & Security: Prometheus, Grafana, OpenTelemetry, ELK, SPIRE, OPA, Vault, mTLS

Experience

- **Confluent** — Remote
  Senior Software Engineer II, Developer Productivity — *April 2025 - Present*
  - **Developer platform strategy**: Co-leading technical direction for the internal developer platform, owning the developer-experience and self-service roadmap and shaping foundational capabilities that help engineering teams provision, deploy, and operate services with less friction.
  - **AI-native developer tooling**: Designed and shipped an AI agent that automates the end-to-end software-vulnerability remediation lifecycle; adopted as the organization's default, it remediated 300+ vulnerabilities in roughly five months. Championed its migration to a new autonomous-workflow platform and drove cross-team design reviews for the next architecture.
  - **AI platform architecture**: Leading evaluation and architecture of the company's AI gateway, a shared control plane for how services access large language models. Drove cross-team technical due diligence and source-level verification that caught a supply-chain compromise in a leading candidate before adoption; an earlier AI-infrastructure evaluation I led avoided more than $500K in annual spend by recommending against an unready deployment.
  - **Reliable internal platforms**: Re-architected a production policy-enforcement service that silently dropped critical configuration-change approvals when synchronous processing exceeded an upstream timeout. Designed an asynchronous, back-pressured architecture with dashboards and alerting, eliminating the silent-failure mode and the on-call toil it caused.
  - **Secure paved paths**: Drove fleet-wide security and reliability hardening by automating service migration to zero-trust mTLS and modern least-privilege secret management, reducing operational toil while shrinking the credential attack surface.
  - **Engineering enablement**: Authored RFCs and technical one-pagers that steered cross-team platform and AI-tooling decisions, taught engineers across the organization to build high-quality AI-assistant plugins, and published reusable plugins to the company's internal marketplace.
- **Angi Inc** — Remote
  Senior Infrastructure Engineer — *Sept 2023 - Mar 2025*
  - **Paved-path platform**: Built foundational components of the Angi One internal platform using Kubernetes controllers, Flux, and ArgoCD, enabling developers to provision applications and cloud services from a single declarative manifest and cutting production delivery time by 50%.
  - **Developer portal experience**: Partnered with application and observability teams to automate Backstage catalog component generation and improve onboarding, service discoverability, and system visibility across the engineering organization.
  - **Documentation as platform capability**: Led automation of documentation workflows from Kubernetes CRDs so platform documentation evolved with releases, reducing manual upkeep and support overhead.
- **Workday** — Victoria, BC, Canada
  Senior Software Development Engineer, DevOps — *May 2023 - Sept 2023*
  Software Development Engineer III, DevOps — *May 2021 - April 2023*
  Software Development Engineer II, DevOps — *May 2019 - April 2021*

- ◦ **Developer self-service**: Implemented policy-based infrastructure workflows using OPA and Atlantis, giving developers safer self-service control over cloud resources while shifting security checks earlier in the delivery lifecycle.
  - ◦ **Cloud delivery platform**: Designed an automated AWS deployment pipeline for microservices across accounts and regions, standardizing service delivery and reducing recurring operations toil by more than four hours per week.
  - ◦ **Kubernetes platform engineering**: Built and operated production infrastructure across AWS and datacenter environments while migrating applications and services onto an internal Kubernetes platform.
  - ◦ **Reusable delivery tooling**: Developed Jenkins pipelines and shared delivery libraries used by multiple applications and services, improving consistency and repeatability across build and deployment workflows.
  - ◦ **Developer ownership**: Spearheaded a cross-team enablement initiative that helped development teams take greater ownership of operating their services; mentored engineers as they grew into larger technical responsibilities.

- **NYU CUSP** Brooklyn, NY, USA
  Associate Research Scientist *May 2015 - May 2019*
  Assistant Research Scientist *June 2014 - May 2015*
  - ◦ **Container platform**: Developed Dockkeeper, a scalable and secure container scheduling and monitoring system using Docker and Prometheus to provision services across physical hosts; eliminated more than $35K per year in VM licensing costs and improved host utilization by over 55%.
  - ◦ **Shared research infrastructure**: Architected the NYU/CUSP Urban Observatory's multi-site physical infrastructure with dense compute and more than half a petabyte of storage, provisioned as multi-user mini-HPC environments using Ansible and Packer.
  - ◦ **Production IoT platform**: Developed a secure, mission-critical IoT platform and CI/CD framework for deploying and operating more than 100 urban noise-monitoring sensors across New York City as part of the NSF-funded SONYC project.

Education

- **NYU Tandon School of Engineering** Brooklyn, NY, USA
  Master of Science in Telecommunication Networks *Aug. 2012 – May. 2014*
  Thesis - CitySynth: Imaging with a Network of Devices

Selected Projects

- **AdguardController**: Developed a Kubernetes controller to automate DNS record management, replacing manual infrastructure changes with a declarative control loop.
- **UCSLHUB**: Developed resilient backend infrastructure using Kubernetes, JupyterHub, and Keycloak for the CUSP UCSL bootcamp, serving hundreds of students each year.
- **Homelab**: Operate a distributed HCI lab with Ansible and Terraform-managed KVM nodes, Ceph storage, Kubernetes, and network segmentation for testing distributed systems and multi-datacenter failure scenarios.

Publications, Teaching Experience and more..

- **View it online**: https://sharmamohit.com/